

GYAN SETU

Chapter-1, Number System Notes

Numbers

1. Natural Number

Definition: Counting numbers starting from 1.

Examples: 1, 2, 3, 4, 5, ...

2. Whole Numbers

Definition: All natural numbers including 0.

Examples: 0, 1, 2, 3, 4, ...

3. Odd Numbers

Definition: Numbers that are not divisible by 2.

Examples: 1, 3, 5, 7, 9, ...

4. Even Numbers

Definition: Numbers that are divisible by 2.

Examples: 0, 2, 4, 6, 8, ...

5. Prime Numbers

Definition: Numbers that have only two factors: 1 and itself.

Examples: 2, 3, 5, 7, 11, 13, ...

Note: *2 is the only even prime number.*

6. Composite Numbers

Definition: Numbers that have more than two factors.

Examples: 4, 6, 8, 9, 10, 12, ...

Note: 1 is neither prime nor composite.

7. Integers

Definition: All positive numbers, negative numbers, and 0 (but no fractions or decimals).

Examples: -3, -2, -1, 0, 1, 2, 3, ...

8. Rational Numbers

Definition: Numbers that can be written in the form p/q , where $q \neq 0$ and p, q are integers.

Examples: $1/2$, $-3/4$, 5, 0.25 ($=1/4$)

9. Irrational Numbers

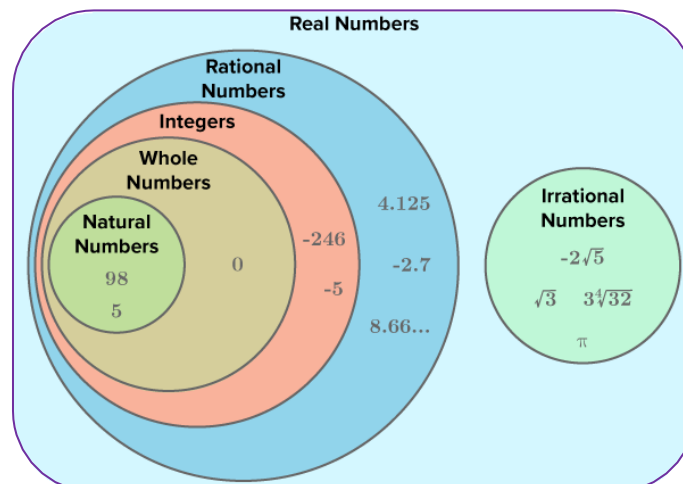
Definition: Numbers that cannot be written as a fraction p/q . Their decimal form is non-repeating and non-terminating.

Examples: $\sqrt{2}$, π , $\sqrt{3}$

10. Real Numbers

Definition: All rational and irrational numbers together are called real numbers.

Examples: -5, 0, $3/4$, $\sqrt{2}$, π



Important Note:

If r is rational and s is irrational, then $r + s$ and $r - s$ are irrational numbers, $(r.s)$ and (r/s) are irrational numbers, $r \neq 0$.

For e.g.:- Let $r = 2$ and $s = \sqrt{2}$

Where r is rational and s is irrational

Therefore, $r + s = 2 + \sqrt{2} \Rightarrow$ Irrational

$r - s = 2 - \sqrt{2} \Rightarrow$ Irrational

$r . s = 2 . \sqrt{2} \Rightarrow$ Irrational

$r / s = 2 / \sqrt{2} \Rightarrow$ Irrational

11. Decimal Numbers

Decimal numbers are numbers that have a **decimal point (.)**, which separates the whole number part from the fractional part.

For example:

- 3.143.143.14
- 7.07.07.0
- 0.333...0.333...0.333...

[1] Terminating

A terminating decimal is a decimal number that ends or *terminates* after a certain number of digits.

Examples:

$0.5 \rightarrow$ stops after 1 decimal place

$1.25 \rightarrow$ stops after 2 decimal places

$\frac{1}{4} \rightarrow 0.25$

These decimals do not go on forever.

Key property: A fraction in the form p/q (in lowest terms) has a terminating decimal if and only if the denominator q has only 2 and/or 5 as its prime factors.

[2] Non-Terminating Decimals

A **non-terminating decimal** never ends — it goes on forever after the decimal point.

There are two types of non-terminating decimals:

a) Recurring (Repeating) Decimals

A recurring decimal has digits that repeat in a pattern endlessly.

Examples:

- $1/3 = 0.333...$ (3 repeats)
- $2/7 = 0.285714285714...$ (pattern "285714" repeats)
- $0.121212...$ (pattern "12" repeats)

These decimals are predictable and have a repeating block of digits.

Notation: A bar is often used to show the repeating part.

Key fact: All recurring decimals are rational numbers because they can be expressed as fractions.

b) Non-Recurring (Non-Repeating) Decimals

A non-recurring decimal goes on forever but without any repeating pattern.

Examples:

- $\pi = 3.1415926535...$ (never ends and doesn't repeat)
- $\sqrt{2} = 1.41421356...$

These decimals are irrational numbers — they cannot be written as fractions of integers.

Type	Ends?	Repeats?	Rational/Irrational
Terminating Decimal	Yes	No	Rational
Non-Terminating Recurring	No	Yes	Rational
Non-Terminating Non-Recurring	No	No	Irrational

Important Identities of Real Number:

- 1) $\sqrt{a \cdot b} = \sqrt{a} \cdot \sqrt{b}$
 - 2) $\sqrt{a / b} = \sqrt{a} / \sqrt{b}$
 - 3) $\sqrt{[(a + b) \cdot (a - b)]} = a - b$
 - 4) $(a + \sqrt{b}) \cdot (a - \sqrt{b}) = a^2 - b$
 - 5) $(\sqrt{a} + \sqrt{b})^2 = a + 2 \cdot \sqrt{a} \cdot \sqrt{b} + b$
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Laws of Exponents:

Let $a > 0$ be a real number and p and q be rational numbers. Then:

- 1) $a^p \cdot a^q = a^{p+q}$
 - 2) $a^p / a^q = a^{p-q}$
 - 3) $(a^p)^q = a^{p \cdot q}$
 - 4) $a^p \cdot b^p = (a \cdot b)^p$
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